

Supplementary Material S1

Dirac-Expanded Bracket Notation for the Controlled- X Gate

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Supplement to “Circuit-Level Density-Matrix Channel-Equivalence Verification of a Five-Qubit Coherent EPR-State Transfer Network with a Two-Component GHZ Resource Family”

Purpose and relation to the main manuscript

This supplementary note contains the extended controlled- X derivation that was formerly included as Appendix B of the main manuscript. It is separated from the article because the result is supporting notation rather than a central contribution of the channel-equivalence theorem. The derivation fixes the computational-basis convention, gives a compact Dirac-expanded representation of $CX_{i,j,q}$, embeds the operator explicitly in an arbitrary q -qubit tensor product, and checks the two- and three-qubit conventions used by the manuscript. The notation is motivated by the Dirac extension of CNOT gates discussed by Yamanashi, Ozawa, and Okazaki [1].

1 Dirac-expanded controlled- X representation

The controlled- X gate acting on a q -qubit register is denoted by

$$\boxed{CX_{i,j,q}}, \quad (S1)$$

where i denotes the control qubit, j the target qubit, and q the total number of qubits, with

$$0 \leq i, j \leq q-1, \quad i \neq j. \quad (S2)$$

The computational-basis ordering used throughout the main manuscript is

$$|q_{q-1} \cdots q_1 q_0\rangle = |q_{q-1}\rangle \otimes \cdots \otimes |q_1\rangle \otimes |q_0\rangle, \quad q_k \in \{0, 1\}. \quad (S3)$$

Circuit wires are displayed from top to bottom in ascending index order $q_0, q_1, \dots, q_{q-1}$. Hence the rightmost tensor factor $|q_0\rangle$ corresponds to the top wire and the leftmost tensor factor $|q_{q-1}\rangle$ to the bottom wire.

For a binary variable $x_k \in \{0, 1\}$, let $\bar{x}_k = 1 - x_k$. The binary power of the Pauli- X operator on qubit k is

$$\boxed{X_k^{x_k} = |0\rangle_k \langle x_k|_k + |1\rangle_k \langle \bar{x}_k|_k}. \quad (S4)$$

Thus

$$X_k^0 = I_k = |0\rangle \langle 0| + |1\rangle \langle 1|, \quad (S5)$$

$$X_k^1 = X_k = |0\rangle \langle 1| + |1\rangle \langle 0|, \quad (S6)$$

and, for $x_k, y_k \in \{0, 1\}$,

$$\boxed{X_k^{x_k} |y_k\rangle = |y_k \oplus x_k\rangle} = \begin{cases} |y_k\rangle, & x_k = 0, \\ |\bar{y}_k\rangle, & x_k = 1. \end{cases} \quad (S7)$$

For $CX_{i,j,q}$, the binary value of the control qubit determines whether the target is acted on by I_j or X_j . Restricting first to the control and target factors gives

$$\sum_{x_i=0}^1 |x_i\rangle_i \langle x_i|_i X_j^{x_i} = |0\rangle_i \langle 0|_i I_j + |1\rangle_i \langle 1|_i X_j. \quad (S8)$$

Identity action on all qubits $k \neq i, j$ is understood at this stage.

Applying Equation (S4) to the target qubit yields

$$X_j^{x_i} = |0\rangle_j \langle x_i|_j + |1\rangle_j \langle \bar{x}_i|_j. \quad (S9)$$

Substitution into Equation (S8) gives the compact Dirac-expanded bracket representation

$$CX_{i,j,q} = \sum_{x_i=0}^1 |x_i\rangle_i \langle x_i|_i \left(|0\rangle_j \langle x_i|_j + |1\rangle_j \langle \bar{x}_i|_j \right), \quad (S10)$$

again with identity action on all remaining qubits.

Expanding the sum gives

$$\begin{aligned} CX_{i,j,q} = & |0\rangle_i |0\rangle_j \langle 0|_i \langle 0|_j + |0\rangle_i |1\rangle_j \langle 0|_i \langle 1|_j \\ & + |1\rangle_i |0\rangle_j \langle 1|_i \langle 1|_j + |1\rangle_i |1\rangle_j \langle 1|_i \langle 0|_j. \end{aligned} \quad (S11)$$

The first two terms correspond to control value 0 and leave the target unchanged, whereas the last two correspond to control value 1 and exchange the target states $|0\rangle_j$ and $|1\rangle_j$.

2 Explicit embedding in a q -qubit tensor product

For each $x_i \in \{0, 1\}$, define

$$A_k^{(x_i)} = \begin{cases} |x_i\rangle_i \langle x_i|_i, & k = i, \\ X_j^{x_i}, & k = j, \\ I_k, & k \neq i, j. \end{cases} \quad (S12)$$

Then, consistently with Equation (S3), the fully embedded operator is

$$CX_{i,j,q} = \sum_{x_i=0}^1 A_{q-1}^{(x_i)} \otimes A_{q-2}^{(x_i)} \otimes \cdots \otimes A_1^{(x_i)} \otimes A_0^{(x_i)}. \quad (S13)$$

Equation (S13) fixes the tensor-factor positions explicitly and remains valid for adjacent or nonadjacent qubits and for either $i < j$ or $i > j$.

For a computational-basis input $|q_{q-1} \cdots q_1 q_0\rangle$, define

$$q'_k = \begin{cases} q_k, & k \neq j, \\ q_j \oplus q_i, & k = j. \end{cases} \quad (S14)$$

Then

$$CX_{i,j,q} |q_{q-1} \cdots q_1 q_0\rangle = |q'_{q-1} \cdots q'_1 q'_0\rangle, \quad (S15)$$

with

$$q'_i = q_i, \quad q'_j = q_j \oplus q_i, \quad (S16)$$

and $q'_k = q_k$ for $k \neq i, j$. Equivalently,

$$X_j^{q_i} |q_j\rangle = |q_j \oplus q_i\rangle. \quad (S17)$$

3 Consistency checks used in the manuscript

For $CX_{0,1,2}$, the tensor ordering is

$$|q_1 q_0\rangle = |q_1\rangle \otimes |q_0\rangle, \quad (\text{S18})$$

and qubit 0 is the control while qubit 1 is the target. Hence

$$\boxed{CX_{0,1,2} |q_1 q_0\rangle = |(q_1 \oplus q_0) q_0\rangle}. \quad (\text{S19})$$

For the two-qubit input used in the main manuscript,

$$CX_{0,1,2} (|y\rangle_1 \otimes [\alpha |0\rangle_0 + (-1)^x \beta |1\rangle_0]) = \alpha |y, 0\rangle + (-1)^x \beta |\bar{y}, 1\rangle. \quad (\text{S20})$$

Thus the controlled- X convention reproduces the encoded state definition used in the manuscript.

As a second check, consider $CX_{0,2,3}$. With

$$|q_2 q_1 q_0\rangle = |q_2\rangle \otimes |q_1\rangle \otimes |q_0\rangle, \quad (\text{S21})$$

qubit 0 is the control, qubit 2 the target, and qubit 1 is unchanged. Therefore

$$\boxed{CX_{0,2,3} |q_2 q_1 q_0\rangle = |(q_2 \oplus q_0) q_1 q_0\rangle}. \quad (\text{S22})$$

Consequently, the control-target expression

$$\boxed{CX_{i,j,q} = \sum_{x_i=0}^1 |x_i\rangle_i \langle x_i|_i X_j^{x_i}} \quad (\text{S23})$$

together with the explicit tensor embedding in Equation (S13) provides a single Dirac-expanded representation valid for arbitrary register size and qubit placement.

References

- [1] Yamanashi, S.; Ozawa, K.; Okazaki, H. On Dirac extension bra-ket notations of CNOT gates. *IEICE Tech. Rep.* **2023**, *122*, CAS2022-61, 1–6. (In Japanese).